Cleaning and Merging Datasets

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Methods

Genetic and phenotypic data were collected by the team of Dr George Perry at Pennsylvania State University with the University of Antananarivo. Dr Jason Hodgson and Dr Rindra Rakotoarivony led this data collection in 2015. These data comprise of 192 individuals from five regions: Merina people from the Central Highlands (n = 36), Betsileo people from the southern Highlands (n = 38), Sakalava and Tsimihety people from the northwestern coast (n = 40), Betsimisaraka people from the eastern coast (n = 38), and people of various ethnic groups from the extreme north in the town of Diego (n = 40). Samples were collected from married couples, and ~690,000 SNPs were typed using the Illumina HumanOmniExpress-24 array.

Two main datasets were created, a Malagasy only dataset and a global dataset combining Malagasy samples with comparative data. The Malagasy dataset was created by merging the data from this study with Malagasy data from (Pierron et al. 2014) (n = 69). For a comparative global dataset, this data was combined with data from southeast Asian and African populations in the (Auton et al. 2015) dataset (n = 882), Bantu African individuals from (**fortes-lima2024?**) (n = 1763), individuals from Borneo from (Brucato et al. 2016) (n = 39), and individuals from the islands of Sumba (n = 30, subsampled at random from the full dataset of 240 individuals) and Timor (n = 7) from (Cox et al. 2016).

Data were cleaned and merged in R with package tidypopgen (Carter and Manica 2025). Only autosomal SNPs were analysed. All SNPs with a missingness rate above 0.01 across all samples were removed. All individuals with a genotype call rate less than 0.9 (missingness exceeding 0.1) were removed. All SNPs with minor allele frequency below a threshold of 0.025 were removed. All SNPs with a Hardy-Weinberg equilibrium test p-value below 0.01 were removed. Close relatives were removed using the KING robust kinship coefficient (Manichaikul et al. 2010), implemented in tidypopgen. One individual was removed from each pair with a kinship coefficient greater than 0.0884, equivalent to removing relatives up to the second degree. After this filtering, the Malagasy dataset consisted of 233 individuals and 483776 SNPs, and the global dataset consisted of 2738 individuals and 215593 SNPs.

SNPs in linkage were then removed by LD clumping (Privé et al. 2018), using tidypopgen. Those with an r2 value greater than 0.2 in a sliding window of 100 kilobases were removed. After pruning, the Malagasy dataset consisted of 182203 SNPs and the global dataset consisted of 126932 SNPs.

Principal components analysis was used to visualise Malagasy individuals compared to global comparative data. Non-malagasy populations were randomly subsampled to a maximum of 30 individuals to avoid bias through uneven sample sizes. This left a total of 1952 individuals.

Unsupervised clustering in fastMIXTURE (Santander, Refoyo Martinez, and Meisner 2024) was run on the full, cleaned dataset of 2738 individuals and 215593 SNPs for 9 values of K (2to 10) and 3 runs per K. Subsequently, we analysed the output of fastmixture using the package tidygenclust (ref), which implements the clumppling algorithm of (Liu, Kopelman, and Rosenberg 2024) in R. We found a single mode for each value of K across 3 replicates.

Admixture f3 statistics were used in the configuration f3(Malagasy; Asian, African) to examine which comparative African and Asian populations look most like the 'source' populations of the Malagasy.

F4 ratio statistics were used in the configurations f4(French,Gorilla;Malagasy,Borneo)/f4(French,Gorrilla;Esan,Borneo) and f4(French,Gorilla;Malagasy,Esan)/f4(French,Gorrilla;Borneo,Esan) were used to examine the proportion

of ancestry derived from Asian vs African sources for each population. These were implemented using the R packages tidypopgen and Admixtools2 (Carter and Manica, 2025; Maier and Patterson, 2024).

ALDER (version 1.03) (Loh et al. 2013) was used to date the timing of admixture across different Malagasy groups. As ALDER requires accurate genetic distances across loci, we used gt_order_loci() to create a subset of the full dataset containing only SNPs with known genetic map positions in correct order. This subset consisted of 215557 SNPs. ALDER was run using the default -binsize of 5×10^{-4} (describing the size, in Morgans, of the bins that SNPs are split up into. Instead of directly calculating distance between each SNP pair, ALDER uses the distance between the bins of each SNP pair.), and the –mindis parameter set to 0.005. Standard error of estimated dates were calculated through leave-one-chromosome-out jackknifing, with the flag -jackknife.

To relate estimated generation times to calendar years, we assumed a generation time of 29 years (refs).

Pairwise Fst distances between malagasy populations were calculated in tidypopgen, using the method of Weir and Cockerham, 1984 (ref).

Bibliography:

Auton, Adam, Gonçalo R. Abecasis, David M. Altshuler, Richard M. Durbin, Gonçalo R. Abecasis, David R. Bentley, Aravinda Chakravarti, et al. 2015. "A Global Reference for Human Genetic Variation." *Nature* 526 (7571): 68–74. https://doi.org/10.1038/nature15393.

Brucato, Nicolas, Pradiptajati Kusuma, Murray P. Cox, Denis Pierron, Gludhug A. Purnomo, Alexander Adelaar, Toomas Kivisild, Thierry Letellier, Herawati Sudoyo, and François-Xavier Ricaut. 2016. "Malagasy Genetic Ancestry Comes from an Historical Malay Trading Post in Southeast Borneo." *Molecular Biology and Evolution* 33 (9): 2396–2400. https://doi.org/10.1093/molbev/msw117.

Carter, Evelyn, and Andrea Manica. 2025. *Tidypopgen: Tidy Population Genetics. R Package Version 0.0.0.9022*. https://evolecolgroup.github.io/tidypopgen.

Cox, Murray P., Georgi Hudjashov, Andre Sim, Olga Savina, Tatiana M. Karafet, Herawati Sudoyo, and J. Stephen Lansing. 2016. "Small Traditional Human Communities Sustain Genomic Diversity over Microgeographic Scales Despite Linguistic Isolation." *Molecular Biology and Evolution* 33 (9): 2273–84. https://doi.org/10.1093/molbev/msw099.

Liu, Xiran, Naama M Kopelman, and Noah A Rosenberg. 2024. "Clumppling: Cluster Matching and Permutation Program with Integer Linear Programming." *Bioinformatics* 40 (1): btad751. https://doi.org/10.1093/bioinformatics/btad751.

Loh, Po-Ru, Mark Lipson, Nick Patterson, Priya Moorjani, Joseph K Pickrell, David Reich, and Bonnie Berger. 2013. "Inferring Admixture Histories of Human Populations Using Linkage Disequilibrium." *Genetics* 193 (4): 1233–54. https://doi.org/10.1534/genetics.112.147330.

Manichaikul, Ani, Josyf C. Mychaleckyj, Stephen S. Rich, Kathy Daly, Michèle Sale, and Wei-Min Chen. 2010. "Robust Relationship Inference in Genome-Wide Association Studies." *Bioinformatics* 26 (22): 2867–73. https://doi.org/10.1093/bioinformatics/btq559.

Pierron, Denis, Harilanto Razafindrazaka, Luca Pagani, François-Xavier Ricaut, Tiago Antao, Mélanie Capredon, Clément Sambo, et al. 2014. "Genome-Wide Evidence of Austronesian–Bantu Admixture and Cultural Reversion in a Hunter-Gatherer Group of Madagascar." *Proceedings of the National Academy of Sciences* 111 (3): 936–41. https://doi.org/10.1073/pnas.1321860111.

Privé, Florian, Hugues Aschard, Andrey Ziyatdinov, and Michael G B Blum. 2018. "Efficient Analysis of Large-Scale Genome-Wide Data with Two r Packages: Bigstatsr and Bigsnpr." *Bioinformatics* 34 (16): 2781–87. https://doi.org/10.1093/bioinformatics/bty185.

Santander, Cindy G., Alba Refoyo Martinez, and Jonas Meisner. 2024. "Faster model-based estimation of ancestry proportions." *Peer Community Journal* 4. https://doi.org/10.24072/pcjournal.503.